

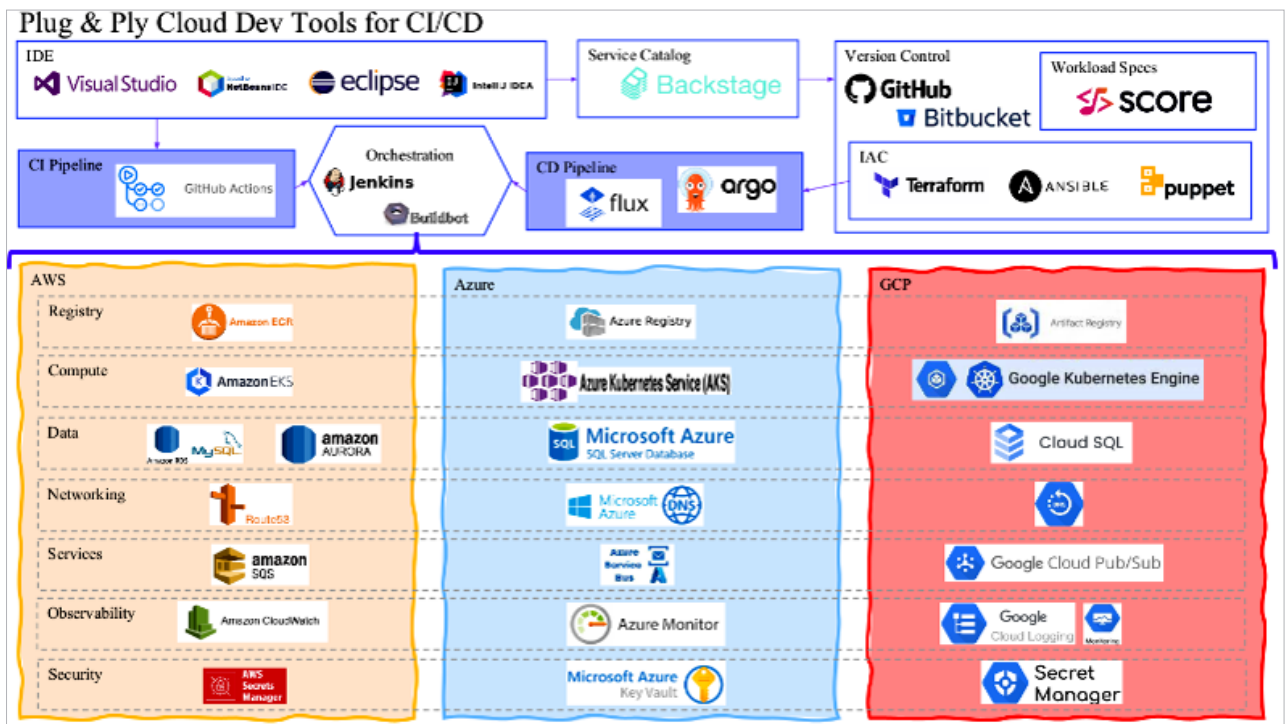


Figure 1: Platform engineering reference architecture

Establish a frequent collaboration route with platform engineers and developers to share feedback on what's working and what's not, allowing for constructive dialogue enabling knowledge sharing while progressing towards the common goal of establishing a robust internal developer platform.

AI-augmented development

The use of AI technologies to develop software, and test and deliver applications, is popular with developers working to design tools to code while adopting AI coding assistants like GitHub CoPilot, Code Whisper, and ChatGPT. Augmented testing tools have made huge progress in addressing the gaps in testing data quality, validation of test results, and right sizing of the testing for the given scope of development work. This enables platform engineers and testing experts to collaborate on incorporating the right mix of tools within the internal developer platforms across CI/CD pipelines. Many standards are evolving, including in the US, where

the Biden administration has released an executive order on the safe, secure, and trustworthy adoption of AI.

GitHub Actions: Powerful workflow automation tool

GitHub Actions is a widely adopted toolset for the automation of build, test, and deployment pipelines as part of platform engineering. The tool is open source, integrates seamlessly with the GitHub repository, works on-premise, and supports cloud-based development across Linux, Windows, and MacOS operating systems. Ease of use has been one of its biggest plus points as compared to the more familiar Jenkins, which has a steep learning curve for setting up on servers initially. The tool is designed for developers to fully utilise in their CI/CD pipeline, and familiarity with YAML helps.

GitHub Actions can be integrated quickly with existing toolsets, allowing for the build, test, deployment, and scaling of containers. It does not offer platform-specific or

granular controls like Azure DevOps, supporting stages defined in the YAML file for creating deployment workflows. This is achieved in GitHub Actions by separating stages into different YAML workflow files, which act as standardisation modularising pipelines for better maintenance and debugging.

Argo: Orchestrating CI/CD workflows on containers

Argo, the container native workflow engine, is designed for orchestrating parallel jobs on Kubernetes. It's a mature tool that is widely adopted in all types of pipelines. It enables developers to orchestrate complex CI/CD pipelines, providing a declarative workflow definition syntax that ensures clarity and conciseness while maintaining scalability and resource optimisation. Although there are other feature-rich tools like Airflow that are more versatile, Argo excels in its simplicity and quick integration for developing reusable workflows.